

Lecture 4 Tutorial — Answers

Q1. Core Concepts

1. A **point** estimate is a single value used to estimate a population parameter.
2. The **confidence** level is (for example, 0.95).
3. For the standard normal distribution, a **critical** value is a z score on the borderline separating significantly low/high values.
4. A **confidence** interval gives a range of plausible values for a population parameter.

Q2. Confidence Interval Interpretation

B. We are 95% confident that the interval produced by this method contains the true value of p .

Q3. Confidence Level and Tails

For a 95% confidence level:

1. $\alpha = 0.05$
2. $\alpha/2 = 0.025$
3. $1 - \alpha/2 = 0.975$

The cumulative area to the left of the positive critical z value is **0.975**.

Q4. Determining Sample Size

The three ingredients are:

1. **Critical z value** (or z score separating an area of $\alpha/2$ in the right tail)
2. **Sample proportion** (if known)
3. **Desired margin of error**

Q5. Estimating a Population Mean

1. $n = 45 \rightarrow$ use **Normal**
2. $n = 18 \rightarrow$ use **Student t**

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Q6. Estimating a Population Variance

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

In this formula, s^2 is the **sample** variance, and σ^2 is the **population** variance.

Q7. Null and Alternative Hypotheses

$$H_0 : \mu_A = \mu_B$$

$$H_a : \mu_A \neq \mu_B$$

Test type: **two-tailed**

Q8. Significance Level and Error Interpretation

1. **T**
2. **F** — A larger α generally makes rejection of H_0 **easier**.
3. **F** — The parameter tested is **not always a mean**; it can also be a variance or a proportion.

Q9. Parameter-to-Distribution Mapping

A-3, B-2, C-1

That is:

- Mean $\rightarrow$ Student distribution (t)
- Variance $\rightarrow$ Chi-square distribution
- Proportion $\rightarrow$ Normal distribution

Q10. Test Type, Direction, and Critical Region

1. $H_a : \mu \neq \mu_0$ **C.** Two-tailed
2. $H_a : \mu < \mu_0$ **A.** Left-tailed
3. $H_a : \mu > \mu_0$ **B.** Right-tailed
4. The lecture states that different test types lead to different selected **critical** values.
 B. z scores (critical values)

Q11. Decision Rule and Conclusion Wording

Use the rule: If $P\text{-value} \leq \alpha$, reject H_0 .

1. $P = 0.031$, $\alpha = 0.05$ **Reject H_0**
2. $P = 0.120$, $\alpha = 0.05$ **Fail to reject H_0**
3. $P = 0.010$, $\alpha = 0.01$ **Reject H_0**
4. $P = 0.0101$, $\alpha = 0.01$ **Fail to reject H_0**

If we fail to reject H_0 , there is **not sufficient** evidence to support / warrant rejection of the claim.

Q12. Two Samples, ANOVA, and Non-parametric Tests

Part A — Two-sample proportion hypotheses

$$H_0 : p_1 = p_2 \iff H_0 : p_1 - p_2 = 0$$

$$H_1 : p_1 \neq p_2 \iff H_a : p_1 - p_2 \neq 0$$

Part B — One-way ANOVA

B. Equality of three or more population means

Part C — ANOVA requirements

The conditions listed in the lecture are:

- A. **A.** Populations are approximately normal
- B. **B.** Populations have the same variance (or standard deviation)
- C. **C.** Samples are simple random samples of quantitative data
- D. **D.** Samples are independent
- E. **F.** Different samples are from populations categorized in only one way

Part D — Non-parametric tests

One advantage:

- They have less rigid requirements and can be applied to a wider variety of situations.

One disadvantage:

- They are not as efficient as parametric tests, so stronger evidence is generally needed to reject H_0 .